YEAR 7 · SCIENCE · MODELS · CHALLENGE HOMEWORK

Why Is Mr Bowen Still Awake?

Half-life with real coffee, a real interview, and carbon dating



Name: _____ **Class:** _____

Date given: _____ **Date due:** _____

How to do this homework

- **Parts A and B:** Mr Bowen's coffee. They teach you the method.
- **Part C is the main task.** You interview an adult in your family. **Do the interview at the weekend.**
- **Part D:** carbon dating.
- You need a **calculator**, a **ruler**, and a **computer or phone**. Answer in **full sentences**.

Remember from the lesson

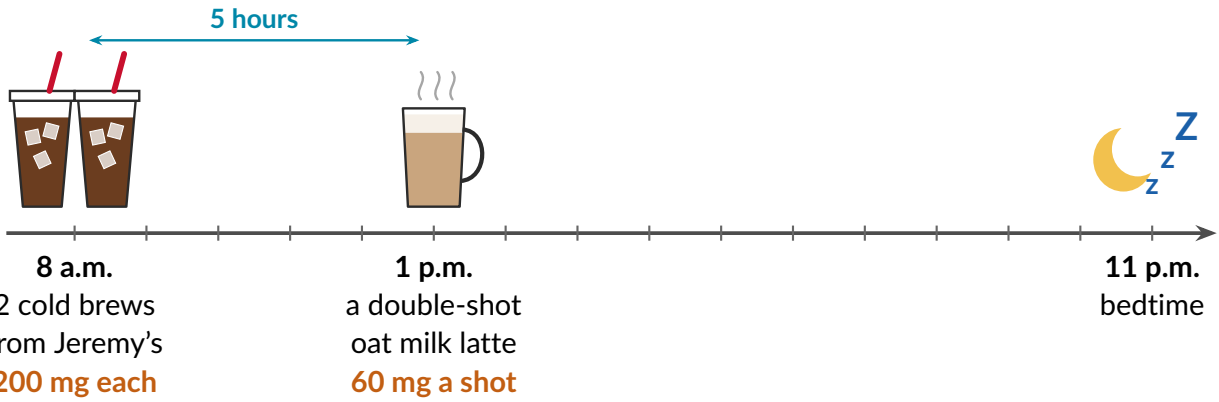
Half-life: the time it takes for half of something to disappear. Caffeine's half-life is about **5 hours**.

Carbon-14: a rare kind of carbon that slowly breaks down into nitrogen. Its half-life is **5 700 years**.

Carbon dating: finding the age of something that was once alive by measuring how much carbon-14 is left. It only works for things that were **alive**, up to about **50 000 years** old.

Part A Mr Bowen's Real Day

This is what Mr Bowen really drinks on a school day.



Word help

caffeine	a chemical in coffee, tea, cola and energy drinks. It keeps you awake.
mg	milligram. 1 000 mg = 1 g. One grain of rice is about 20 mg.
cold brew	strong coffee made slowly with cold water
double shot	two shots. A **shot** is a small cup of very strong coffee.
latte	coffee with a lot of milk. **Oat milk** is made from oats.

A1 The drinks

(a) The cold brews: $2 \times 200 =$ _____ mg (b) The latte: $2 \times 60 =$ _____ mg

(c) Does the **oat milk** add any caffeine? ☐ yes ☐ no

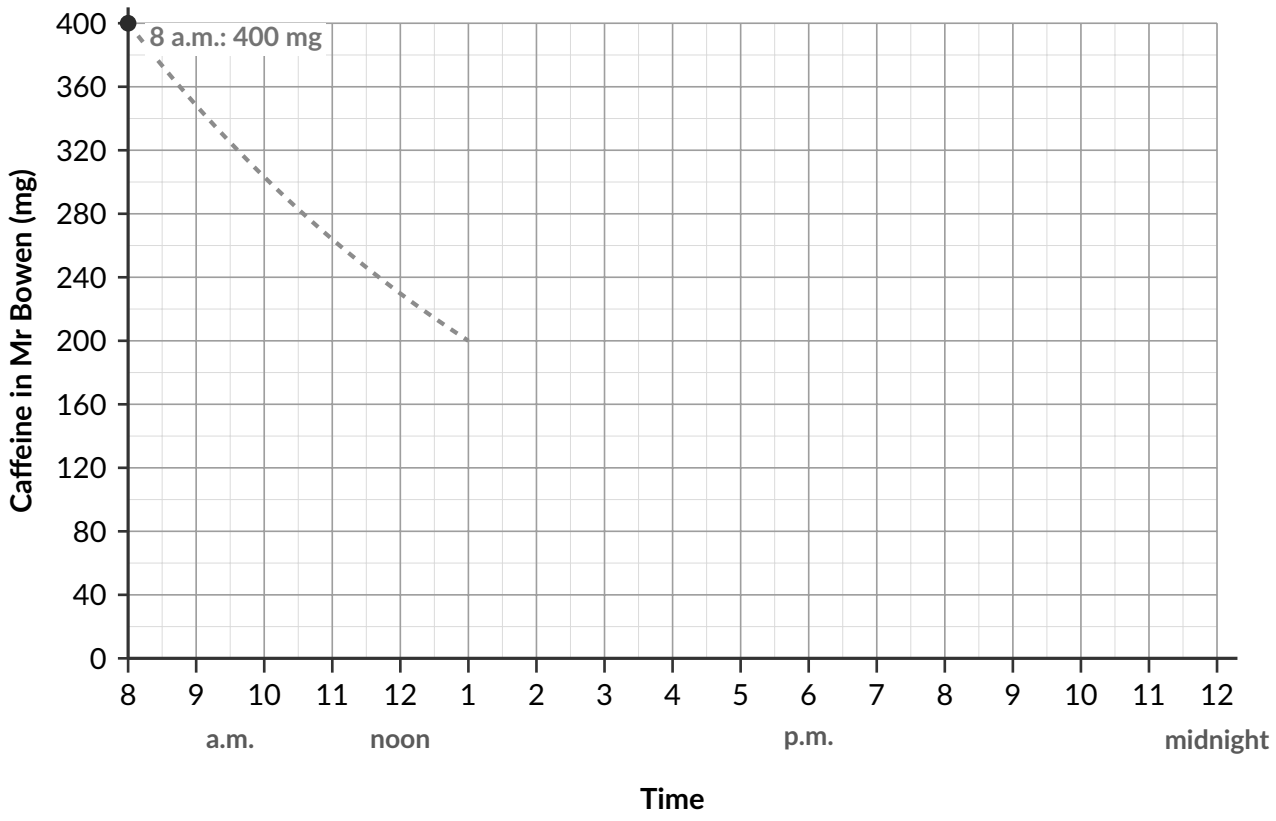
A2 Half-life steps

Fill in the boxes. Every 5 hours, **halve** (cut in half). When he drinks, **add**.

8 a.m. 400 mg → 5 hours ÷ 2 → **1 p.m.** _____ mg → latte + → **1 p.m.** after the latte _____ mg → 5 hours ÷ 2 → **6 p.m.** _____ mg → 5 hours ÷ 2 → **11 p.m.** bedtime _____ mg

A3 Draw the graph

Put a dot for every box in A2. The morning is drawn for you. At 1 p.m., draw a line **straight up**. Between the dots, draw a **smooth curve**.



A4 Read the graph

(a) At 8 p.m. Mr Bowen is marking your homework. Go **up** from 8 p.m. to your curve, then **across**. How much caffeine is in him? _____ mg

(b) How much caffeine is in him at bedtime, 11 p.m.? _____ mg

(c) Health experts say an adult should have **no more than 400 mg** of caffeine in a day. How much did Mr Bowen drink?

_____ + _____ = _____ mg Is that more than 400 mg? _____

(d) Tomorrow he wants **exactly 400 mg**. Which **one** drink should he not buy?

He should not buy _____.

A5 The title question

Why is Mr Bowen still awake at 11 p.m.? Use a number, and compare it with a cup of coffee (about 100 mg).

At 11 p.m. Mr Bowen still has …

Part B Hour by Hour

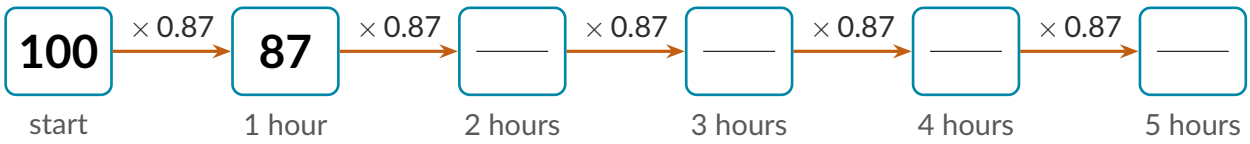
The hour-by-hour rule: it works for any time

Every hour: × 0.87. Round to a whole number. **When they drink: + the drink.**

Example: Mr Bowen at 1 p.m. 230 × 0.87 = 200, then + 120 for the latte = **320** mg.

B1 Why 0.87?

Start with 100 mg. Multiply by 0.87 every hour. Round each answer.



After 5 hours, about how much is left? _____ That is one _____.

B2 Check it with Mr Bowen

Use the rule from 1 p.m. to 8 p.m. (He drinks nothing in this time.)

1 p.m.	2 p.m.	3 p.m.	4 p.m.	5 p.m.	6 p.m.	7 p.m.	8 p.m.
320							

Is your 6 p.m. number about the same as in A2? ☐ yes ☐ no

Is your 8 p.m. number about the same as in A4(a)? ☐ yes ☐ no

When different methods agree, scientists _____ the answer.

Part C Interview a Caffeine Drinker

THE MAIN TASK

What you will do

Interview an adult in your family who drinks caffeine: a parent, a grandparent, an aunt, an uncle. Ask about coffee, tea, soda (like Coca-Cola) or **energy drinks like Sting**. Then **estimate** how much caffeine is in their body at bedtime.

Nobody at home drinks caffeine? Ask another adult you know. You can do the interview in **any language**. Write in **English**.

C1 The interview

Ask about **yesterday**, or another normal day.

1. "Can I interview you for my homework?"
2. "What did you drink that has caffeine?"
3. "What time did you drink it?"
4. "How big was it? A cup, a can, a bottle?"
5. "What brand was it?"
6. "What time did you go to bed?"

I interviewed my _____. **They went to bed at** _____.

Write their drinks here. Round each time to the **nearest hour**: 7:40 → 8:00, 10:20 → 10:00. Fill in the last two columns in C2.

Drink what, and how big	**Time**	**Rounded time**	**Caffeine (mg)**	**Where I found it**
Example: 1 can of cola, 330 ml	3:40 p.m.	4 p.m.	32	the company's website

C2 Find the caffeine

Where to look

1. **The label** on the can or bottle. Look for the word **caffeine**. It often says how many mg are in **100 ml**.
2. **The company's website.**
3. **Search:** *caffeine in [the drink] mg*, for example *caffeine in Sting 330 ml*.

Milk, sugar, ice and condensed milk: **0 mg**.

Two numbers, like 50 to 80 mg? Use the **middle**: 65 mg.



Reading a label. This bottle has 20 mg in every 100 ml. It holds 330 ml.

330 ÷ 100 = 3.3 3.3 × 20 = **66 mg** in the whole bottle.

(a) **10 mg in 100 ml**, a **330 ml** can: _____ × 10 = _____ mg

(b) **30 mg in 100 ml**, a **250 ml** can: _____ × 30 = _____ mg

Now look up every drink in C1. Fill in the **Caffeine** and **Where I found it** columns.

C3 Hour by hour

Use the rule from Part B and the **rounded** times. Row 1 is their **first** drink. Each row is **one hour later**. Stop at **bedtime**.

Start here

Time	Last hour × 0.87	+ Drink	Now (mg)

Then carry on here

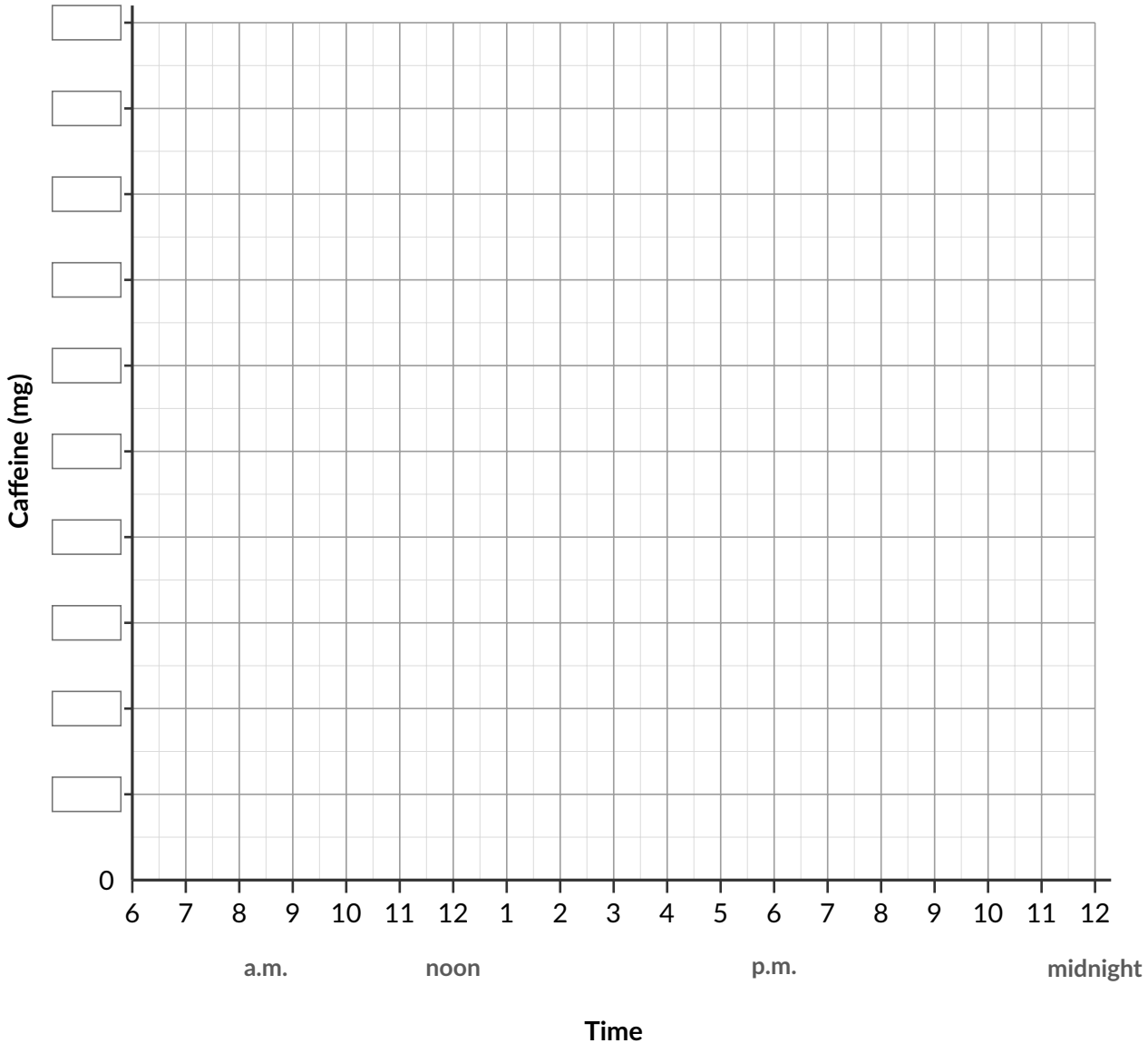
Time	Last hour × 0.87	+ Drink	Now (mg)

C4 Draw their graph

Step 1. Find the biggest number in your table. Choose one big square from this table, and write the numbers in the boxes on the graph.

Biggest number	under 100	under 200	under 300	under 400	under 600
One big square =	10 mg	20 mg	30 mg	40 mg	60 mg

Step 2. Put a dot for every row of your table. At a drink, go **straight up**. Between the dots, draw a **smooth curve**.



C5 Your results

(a) At bedtime (_____), my _____ had about _____ mg of caffeine in their body.

(b) Compare with Mr Bowen (80 mg at bedtime).

My _____ had more / less caffeine at bedtime than Mr Bowen, because …

(c) **Challenge.** They want less caffeine at bedtime. Give them **one** piece of advice. Use a number from your table.

I think they should … because …

(d) Your model is an estimate. Give **one** reason your bedtime number might be wrong. *Clue:* caffeine's half-life is not 5 hours for everybody. It can be from about 1.5 to 9.5 hours.

My bedtime number might be wrong because …

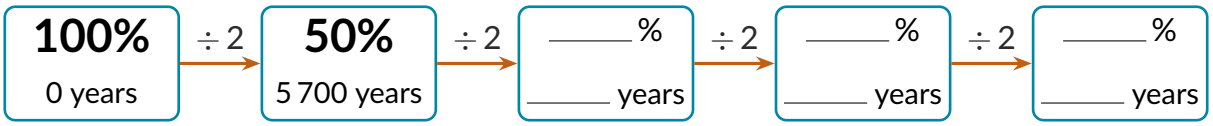
Before you hand it in: tick each one

- ☐ Every drink in C1 has a time, the caffeine in mg, and where I found it.
- ☐ My C3 table goes up **one hour** a row and stops at bedtime.
- ☐ Each drink is added **once**, in its own hour.
- ☐ My graph goes **straight up** at every drink.
- ☐ My bedtime number is the **same** in C3, C4 and C5(a).

Part D Carbon Dating

D1 The halving chain

Caffeine halves every 5 hours. Carbon-14 halves every **5 700 years**. Fill in the boxes.



D2 The museum

Scientists measured the carbon-14 left in six objects. Use D1. **Careful:** carbon dating does not work for everything.

Object	Was it ever alive?	Carbon-14 left	How many half-lives?	Age (years)
A wooden boat from a river	yes	50%	1	5 700
A bear bone from a cave		12.5%		
Charcoal (burnt wood)		6.25%		
A cotton T-shirt, made this year		100%		
A stone axe		—		
A *Tyrannosaurus rex* tooth		0%		

Word help: cotton comes from a plant · an **axe** is a tool for cutting wood

D3 Same model, different clock

While Mr Bowen is alive, **coffee** tops up his caffeine and **food** tops up his carbon-14. What is the **same** about caffeine and carbon-14? What is **different**? Use the word **half-life**.

They are the same because … The difference is …